FAT16 Tables (for class reference)

Basic Layout

This it the most common layout I'm going to use, but it's configurable (see boot sector).

Section	Boot Sector	FAT1	FAT2	Root Directory	Data
Sector(s)	0	1–32	33–64	65–96	97–8195
Cluster(s)	—	—	—	—	2–8100

Boot Sector

This is the first 512 bytes of the disk. It tells us how to interpret the rest.

Offset	Size	Field	Typically
0	3	Jump Instruction	`EB 3C 90`
3	8	OEM Name	`mkfs.fat`
11	2	Bytes per sector	`0x200 (512)`
13	1	Sectors per cluster	1
14	2	Reserved sectors	1 (boot sector)
16	1	Number of FATs	2
17	2	Number of Root Entries	`0x200 (512)`
19	2	Total Sectors (16-bit)	`0x2000 (8192)`
21	1	Media descriptor	`0xF8`
22	2	Sectors per FAT	`0x0020 (32)`
24	2	Sectors per track	`0x0020 (32)`
26	2	Number of heads	2
28	4	Hidden sectors	0
32	4	Total sectors (32-bit)	0
36	1	Drive number	
37	1	Reserved	
38	1	Extended boot signature	
39	4	Volume ID	
43	11	Volume label	
54	8	Filesystem type	FAT16
62	—	Boot code / paadding	
510	2	Boot signature	`55 AA`

FAT entries

Value	Meaning
`0x0000`	Cluster is free
`0x0002–0xffef`	Cluster in use. Specifies the next cluster in the file
`0xfff0–0xfff6`	reserved (shouldn't be there)
`0xfff7`	Bad cluster
`0xfff8–0xffff`	Cluster in use. Last cluster in the file.

Directory Entries

Starts at (offset)	Length (in bytes)	Description
0	8	Filename
8	3	Extension
11	1	Attributes (see below)
12	2	Reserved
14	2	Creation Time
16	2	Creation Date
18	2	Last Access Date
22	2	Last Write Time
24	2	Last Write Date
26	2	First Logical Cluster
28	4	File Size (in bytes)

Bit	Attribute
0	Read Only
1	Hidden
2	System
3	Volume Label
4	Directory
5	Archive
6	—
7	—